Stop paying for freight the plan never asked for

Northstar Frozen Foods · Out-of-route freight leakage · Business case & solution

Prepared for: VP Supply Chain (executive sponsor) and Director, Transportation & Network Planning (domain owner)

Northstar Frozen Foods is a fictional frozen bakery & foodservice manufacturer; all figures come from a synthetic dataset built for this prototype and computed live on Databricks (see `evidence/business_kpis.md`).

1 · The outcome: $5.2M a year back from freight, starting with 40 lanes

$23.6M	gross freight leakage in the last 12 months: out-of-route (OOR) moves that cost more than the planned lane (**30.5% of $77.3M freight spend**)
$10.3M / yr	ML-expected leakage on the **40 lanes** now in the planner queue (4% of the 1,068 lanes with OOR, ≈44% of the $1.97M/month leakage run-rate)
$5.2M / yr	target recovery at 50% capture of the queue → **cost per case $1.656 → $1.545 (–6.7%)**, freight-to-sales **1.86% → 1.74%**
Daily, not month-end	leakage is flagged the morning after the load moves, with a named owner and an action, instead of a GL variance 4–6 weeks later

The ask: a 90-day pilot on the Drummondville, McAllen and Joliet origins (7 of the top 10 queued lanes), with success measured against the KPIs on slide 9.

2 · The problem: a third of freight runs off-plan, and nobody sees which third costs money

- **35% of shipped cases** (53% of freight spend) moved on a lane that was **not in that month's MRP plan** for that SKU.
- Not all of it is bad: **56.6%** of OOR lane-SKU-months cost more than plan (avg **+$9.03 per case**), and the rest were opportunistically cheaper (–$6.9M). Treating all OOR the same buries planners in noise.
- The signal is scattered across three systems that don't agree with each other:
- the **TMS** carries legacy ECC plant codes on ~30% of stops and strips leading zeros from ~40% of SKUs,
- Canadian freight is invoiced in **CAD**,
- the **MRP** plan is a monthly snapshot in procurement direction,
- and **Finance** sees only a GL freight variance at month end.
- **Result today:** leakage is discovered late, attributed to "freight rates", and argued about in spreadsheets. The real root cause is often a stale plan.

3 · For the executive sponsor (VP Supply Chain / CFO): a P&L lever you can bank

Capture of queued leakage	Annual savings	Cost per case	Freight % of sales
Today	–	$1.656	1.86%
25% (conservative)	**$2.6M**	$1.601	1.80%
50% (pilot target)	**$5.2M**	**$1.545**	**1.74%**
75% (stretch)	$7.8M	$1.490	1.68%

- That's **$2.6–7.8M of EBITDA** from the first 40 lanes alone. The full network leakage pool is **$23.6M gross / $16.7M net** a year.
- **No new freight contracts, no new headcount:** it uses the planners and supply planners you already have, pointed at the right lanes.
- **Governed by design:** carrier contract rates are column-masked, gold tables are certified and tagged, and every recommendation and decision is auditable.

4 · For the domain owner (Director, Transportation & Network Planning): from 1,068 lanes to a 40-lane queue

- **Prioritised:** an ML model ranks every OOR lane by P(costly) × expected extra $/case × volume. The **top 20% of lanes hold 54%** of out-of-time leakage, so effort goes where the dollars are.
- **Explained:** for each lane, generative AI writes the action, the owner and the $ at stake. It uses only that lane's facts, for example:

  > *"[REROUTE_TO_PLAN] Reroute the 116 out-of-route shipments (142,459 cases) from Joliet Plant to Columbus DC instead of Atlanta DC, using the planned alternative lane 2507-2551 at $0.47/case. This will eliminate the $54,174 realized leakage caused by shipping 573.9 miles instead of the planned 346.2 miles." (verbatim model output, lane #4)*

- **Actionable in one click:** Approve / Reject / Snooze in the Ops Console. Each decision is written to the operational database with who, when and the $ at stake.
- **Insight the data surfaced: 66% of queued $ is a planning problem, not a carrier problem.** 26 of the 40 lanes are `FIX_MRP_PLAN` ($569K/month), owned by supply planning. The cross-border Drummondville plant sits on 3 of the top 5 lanes.

5 · What the planner sees

1. **Ops Console** (`/ops`): KPI strip with queue leakage $/month, realised leakage, high-risk lanes, lanes actioned and $ addressed. Below it, the ranked lane queue with risk tier, P(costly), expected $/month, the GenAI action and owner, and Approve / Reject / Snooze.
2. **3D network map**: plants, DCs and partners with planned vs out-of-route lanes, drill-downs (costly / cheaper / loops / no baseline), route reconstruction per SKU, and sales & stock overlays.
3. **Ask Genie**: *"Which 10 lanes have the highest expected monthly leakage?"* or *"Which product families drive the most costly out-of-route moves?"* returns an answer, the SQL behind it and the table, in seconds.

A morning in the life: the planner opens the queue (served in ~250 ms), approves 5 reroutes and sends 3 plan fixes to supply planning. The KPI strip shows $ addressed, and Finance sees the same number.

6 · Why you can trust the numbers

- **Model quality:** validated **out-of-time**, trained on 2025 and tested on Jan–Apr 2026. **AUC 0.917, PR-AUC 0.932, precision 0.82, recall 0.84.** Lane distance vs plan and the planned baseline $/case are the top drivers, which are physical, explainable signals. Realised invoice cost is never used as a feature.
- **Data quality:** 8 pipeline expectations (valid plant codes and coordinates, non-negative cost, valid GL period, lane classification) are enforced on every run, with **0 rows dropped**. Code mismatches are fixed at the source and not in spreadsheets.
- **Grounded AI:** the LLM sees only one lane's facts and must return a fixed JSON schema (action type, recommendation, owner). Outputs are stored, versioned and reviewable.
- **Natural-language answers checked against SQL:** Genie's answers to benchmark questions were compared with hand-written SQL. **2/2 PASS** (high-risk lane count; 2026 leakage $8.49M).

7 · How it works on Databricks: one governed flow, not six tools

Stage	What it does for the business
Lakeflow	Ingests raw ERP, MRP and TMS extracts (16 feeds) with Auto Loader. The declarative bronze → silver → gold pipeline fixes plant codes, SKU keys and currency, and classifies each lane × SKU × month as planned / out-of-route / theoretical
Unity Catalog	One catalog for every layer: governed tags, a column mask on carrier contract rates, least-privilege grants, data-quality results and lineage
ML (MLflow + UC model registry)	OOR cost-risk model `oor_cost_risk@champion` scores every out-of-route move and rolls up a lane watchlist
Gen AI (AI Functions → Claude)	`ai_query` writes the action, owner and $ rationale for the top-risk lanes
Lakebase	Serverless Postgres serves the queue in milliseconds via synced tables and stores planner decisions transactionally
Genie Agent	Natural-language Q&A over gold, ML and GenAI tables, with curated instructions and benchmarks
Databricks App	Map + Genie + Ops Console in one secure app. Users sign in with workspace identity, and the app runs as its own service principal

The whole flow runs as **one scheduled job** (generate/ingest → transform → routes → govern → ML → GenAI → Lakebase sync → Genie checks), last run **SUCCESS** in about 12 minutes on serverless compute.

8 · Rollout plan: 30 / 60 / 90 days

	Scope	Exit criteria
Days 0–30	Connect production ERP/MRP/TMS extracts, re-baseline leakage, train on real history, onboard 3 pilot origins	Leakage baseline signed off by Finance; model AUC ≥ 0.85 out-of-time
Days 31–60	Daily queue live for pilot planners; supply-planning workflow for `FIX_MRP_PLAN`; weekly review with Finance	≥ 80% of queued lanes actioned within 7 days
Days 61–90	Measure realised vs expected savings on actioned lanes; extend to all origins; add a carrier-rate renegotiation feed	≥ 25% reduction in leakage on pilot lanes vs baseline; go/no-go for network rollout

9 · KPIs we will hold ourselves to

KPI	Owner	Baseline (synthetic TTM)	90-day target
Gross OOR leakage $ / month (pilot lanes)	Dir. Transportation	queue: $862K expected / month	–25% to –50%
Freight cost per case	VP Supply Chain	$1.656	$1.545 at 50% capture
Freight % of net sales	CFO	1.86%	1.74%
OOR share of shipped cases	Dir. Transportation	35.0%	–5 pts on pilot origins
Queue actioned within 7 days	Transportation planners	n/a (no queue today)	≥ 80%
Time to detect a costly OOR move	Finance + Logistics	month-end GL (4–6 weeks)	next day

Appendix · Data & assumptions

- Synthetic network: 55 nodes (8 plants, 16 DCs, 31 fulfillment/partner/outlet nodes), ~2,100 SKUs, 16 months (2025-01 to 2026-04), 70.6K loads, 64.1M cases. Generated with `numpy` seed 42. No customer data.
- "Leakage" = (actual $/case – planned baseline $/case) × cases, for OOR moves above baseline. The baseline cascades lane×SKU → SKU → lane → destination averages.
- "Expected leakage" = ML-predicted extra $/case (floored at 0) × cases, averaged per month over the trailing 3 months (2026-02 to 2026-04).
- Savings scenarios apply a capture rate to the 40-lane queue only. Network-wide upside is larger and not counted.
- All figures and the SQL that produced them: `evidence/business_kpis.md`. Model metrics: `evidence/notebook_runs/ml_oor_cost_risk.md`.